Medical Transcription Classification using Text Mining and Machine Learning

A Dissertation submitted for the degree of

Master of Science in

Data Analytics

Submitted by

Sidharth Vengathattil

P2749578

Supervised by

Aisha Badawi

De Montfort University

Leicester, United Kingdom

September 2023

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to my supervisor, Professor Aisha Badawi, for her invaluable guidance, feedback, and support throughout my dissertation research. Her immense knowledge and experience were instrumental in helping me formulate my research questions, methodology, and analysis. I am deeply grateful for her dedication and encouragement which motivated me to keep working during the challenges of this project.

Contents

ACKNOWLEDGEMENT 2
TABLE OF FIGURES 5
ABSTRACT 6
1. INTRODUCTION 6
1.1. Background 6
1.2. Research Problem 7
1.3. Research Questions 7
1.4. Research Aim and Objectives 8
1.5. Significance and Contribution of the Study 9
2. LITERATURE REVIEW 10
2.1. Introduction 10
2.2. What is Medical Transcription? 10
2.3. Need of medical transcription and its impact on healthcare sector: 11
2.4. What is text mining? 12
2.5. Process of text mining: 12
2.5.1. Entity Extraction: 12
2.5.2. Tokenization: 14
2.5.3. Vectorisation: 14
2.5.4. Dimensionality reduction: 16
2.6. Text mining in healthcare sector: 16
2.7. Class imbalance in medical dataset: 17
2.8. Use of machine learning to classify medical data 18
2.9. Gap of Knowledge: 18
2.10. Conclusion: 19
3. METHODOLOGY 20
3.1. Problem definition 20
3.2. Data collection 20
3.3. Tools and techniques 20
3.4. Load dataset 22
3.5. Understanding the data 22
3.6. Data cleaning and preparation 24
3.6.1. Cleaning: 24
3.6.2. Lemmatization: 25

3.6.3. Entity extraction.....25
3.6.4. Exploratory Data Analysis.....26
3.6.5. Filtration.....28
3.6.6. Tokenization.....29
3.6.7. part of speech tagging.....30
3.7. Feature engineering.....32
3.7.1. Vectorisation.....32
3.7.2. Dimensionality reduction.....33
3.8. Class imbalance.....34
3.9. Model building.....36
3.9.1. Random Forest:.....36
3.9.2. Logistic Regression:.....36
4. RESULT AND FINDINGS.....38
4.1. Model Comparison.....38
4.2. Research questions and findings:.....39
5. CONCLUSION.....42
BIBLIOGRAPHY:.....43
APPENDIX.....46

TABLE OF FIGURES

Figure 1: NER workflow 13
Figure 2: Loaded Dataset 22
Figure 3: list of medical specialties 22
Figure 4: Bar chart showing count of medical specialities 23
Figure 5: Transcription as in the dataset before cleaning 24
Figure 6: Cleaned text 25
Figure 7: Lemmatized text 25
Figure 8: Table showing average similarity of words in each speciality 27
Figure 9: Heat map showing correlation of words between medical specialties 28
Figure 10: Remaining specialties after filtration 29
Figure 11: Bar chart showing the frequency of pos tags in the entire transcription 31
Figure 12: t-SNE plot of vectorised data 33
Figure 14: t-SNE plot after PCA 34
Figure 15:Count of specialties before and after applying SMOTE 35
Figure 16: Original class distribution and smote generated class distribution 35
Figure 17:Flow chart of entire process involved in the process 37
Figure 18: Classification report of logistic regression 38
Figure 19: Classification report of Random Forest 38

ABSTRACT

The rapid evolution of healthcare demands innovative solutions to improve medical documentation and decision-making processes. This analysis explores the domain of medical transcription, employing text mining and machine learning techniques for the classification of medical transcription. By extracting useful information from unstructured text, the developed system classify categorizes transcriptions into relevant medical specialties, enabling efficient data management and informed clinical decisions. Through thorough methods and evaluation, this study contributes to the combination of healthcare and technology, paving the way for more accurate, accessible, and timely medical documentation practices.

1. INTRODUCTION

In the healthcare sector, accurate and efficient documentation is paramount. Medical transcription, the process of converting spoken medical reports into written text, plays a crucial role in capturing patient information, diagnoses, treatments, and other essential medical details. However, as the amount of medical data being produced grows and the need for quick and accurate categorization of transcribed texts rises, conventional manual approaches are unable to keep in pace with the needs.

Advancements in technology have opened new ways for revolutionizing the field of medical transcription. This project focuses on using the power of text mining and machine learning techniques to analyse the accuracy and possibility of classification of medical transcripts. By automating the classification of transcribed content, healthcare professionals can enhance their ability to access, interpret, and act upon patient data efficiently, thereby improving patient care and operational efficiency.

1.1. Background

Medical transcription is a critical component of healthcare documentation that involves the conversion of spoken medical reports into written text format. This process serves as a fundamental bridge between healthcare professionals and accurate medical recordkeeping. The resulting transcribed documents encompass a wide range of medical information, including patient histories, diagnoses, treatments, procedures, and other essential clinical details.

However, the conventional practice of manual medical transcription faces several challenges that have caused the need for innovative solutions. As the volume of medical data generated continues to increase, traditional manual transcription methods are no more effective, leading to delays in data availability and decision-making. Moreover, the complexities of medical language and terminologies needs a high level of expertise from transcriptionists to ensure the accuracy and consistency of transcribed documents.

Text mining, which involves the automated extraction of meaningful patterns and insights from unstructured text data, can streamline the processing of medical transcriptions. When combined with machine learning algorithms, text mining enables the development of intelligent models capable of automatically categorizing and classifying medical texts based on their content.

1.2. Research Problem

The challenges in medical transcription are evident in situations like emergency care. Swift, accurate transcription is vital for time-sensitive medical information. Errors can critically impact patient care. Medical language with specialized terms adds complexity. Expertise is needed for accurate transcription and classification. Mistakes could lead to misunderstandings among healthcare providers and affect patient outcomes.

Handling the high volume of medical data generated daily is another challenge. Efficient categorization is crucial for an organized medical record system. Accuracy is paramount in transcribed documents. Even minor errors can result in incorrect diagnoses, treatments, or procedures, risking patient safety.

Text mining and machine learning offer solutions where these technologies automate categorizing medical transcriptions based on content. By analysing language patterns and specialized terms, they distinguish medical specialties, procedures, and diagnoses. This integration has far-reaching benefits.

Automation enhances accuracy by reducing errors in manual transcription. It also saves time and resources, letting healthcare professionals focus on complex tasks. Real-time insights expedite clinical decision-making and improve patient outcomes. Automated systems scale to handle large medical data volumes, ensuring consistent and efficient classification.

1.3. Research Questions

What is the comparative performance of different machine learning algorithms in classifying medical transcriptions into their respective specialties, and how does this performance vary based on the pre-processing steps applied?

How does text mining impact medical data analysis and contribute to simplifying the process?

What are the methods to improve the accuracy of classification?

1.4. Research Aim and Objectives

Aim:

By delving into the capabilities of text mining and machine learning, this project aims to explore their effectiveness in accurately categorizing medical transcriptions. The focus is on analysing the feasibility of classification using these approaches and assessing accuracy through machine learning methods.

The project involves pre-processing of the dataset, application of text mining, use of machine learning, evaluation, and visualization. Evaluation metrics such as accuracy and classification report are used to assess the performance and thereby comparison of the model.

Objectives:

The objective of this project is to assess the viability of employing text mining and machine learning for medical transcription classification. The emphasis lies in understanding the potential of these techniques to categorize medical transcriptions accurately and subsequently evaluating their precision through machine learning algorithms.

The project centres around a dataset of medical transcriptions and their corresponding medical specialties. The project applies text mining and entity extraction techniques, using Python code, to uncover valuable insights from the transcriptions. Additionally, the project employs machine learning algorithms to evaluate the precision of the outcomes.

To achieve this aim, the project is guided by the following specific objectives:

- **Text Pre-processing and Feature Extraction:**

The project seeks to implement a range of text pre-processing techniques, including lemmatization, and utilize the scispaCy model for entity extraction. Additionally, vectorization and dimensionality reduction will be employed as part of the feature engineering process. Subsequently, addressing the class imbalance will be a priority. It also aims to investigate the correlation between different medical specialties. This involves examining how medical transcriptions from various specialties relate to each other based on the content and context of the text.

- **Algorithm Selection and Implementation:**

The project will delve into the assessment of accuracy through the application of major classification machine learning algorithms. A comparative analysis will be conducted to determine the optimal algorithm for classification considering metrics like accuracy and F1-score.

- **Impact Assessment**

Assess the impact of the text mining in healthcare documentation processes.

Discuss future directions for enhancing the system's capabilities to accurately classify transcription data.

By addressing these objectives, this research aims to analyse the contribution in advancement of medical transcription through the integration of text mining and machine learning technologies.

1.5. Significance and Contribution of the Study

This research project on medical transcription classification using text mining and machine learning carries substantial significance within the domains of healthcare documentation, technology, and patient care. Through its innovative approach, the study offers valuable contributions that have the potential to impact various aspects of healthcare and information management.

1. Advancement of Healthcare Documentation:

By seamlessly categorizing transcribed content, the system can ensure the accuracy, consistency, and accessibility of patient information. This advancement aligns with the industry's ongoing pursuit of streamlined data management processes ultimately leading to improved patient care and efficient healthcare operations.

2. Enhanced Accuracy and Efficiency:

The implementation of text mining and machine learning algorithms introduces a new level of accuracy to medical transcription classification. By leveraging algorithms capable of understanding complexities and medical terminology, the system can outperform manual classification methods, reducing the likelihood of human errors and mitigating the risks associated with misclassified medical information.

3. Optimization of Clinical Decision-Making:

Access to well-organized and accurately categorized medical transcripts empowers healthcare professionals with timely and relevant information for clinical decision-making. As a result, the system contributes to optimized patient care, enabling medical practitioners to base their decisions on comprehensive and categorized medical records. This aspect has the potential to positively influence treatment plans, diagnosis accuracy and overall patient outcomes.

4. Scalability and Adaptability:

Scalability ensures that the model can handle the increasing volume of medical data generated by health sector. The machine learning algorithms can adapt to new medical contexts and terminologies ensuring that the classification accuracy remains high even in the face of evolving medical practices. This adaptability is crucial for accommodating emerging medical specialties and terminology changes.

In summation, this research project's significance lies in its ability to combine technology with healthcare's vital documentation processes. It also analyses the potential to enhance accuracy, efficiency, and decision-making represents a meaningful contribution to both the healthcare and technology sectors, with extensive implications for patient care, medical record management, and the broader landscape of medical research and practice.

2. LITERATURE REVIEW

2.1. Introduction

In the realm of healthcare, effective information management is crucial. This literature review delves into medical transcription's history, role, and its need for accuracy. It then navigates to the realm of text mining, a potent tool for deriving insights from textual data. Techniques like tokenization, entity extraction, vectorisation, PCA analysis and SMOTE are discussed in the context of text mining, setting the stage for their application in the research project by reviewing various research papers.

Transitioning to healthcare, the review explores how text mining aids in pattern discovery, impacting patient care and decision-making. It addresses class imbalance within medical datasets, a challenge that affects machine learning models. Lastly, examines the fusion of machine learning and medical transcription classification, aligning text mining, and advanced algorithms.

This review encompasses medical transcription's evolution, the potency of text mining, and its synergy with machine learning for medical data classification. As healthcare advances further into the digital era, these techniques are at the forefront of revolutionising information utilisation, patient care, and medical research.

2.2. What is Medical Transcription?

Medical transcription is the process of converting spoken or dictated medical reports, notes, and records into written or electronic text format. It involves transcribing the verbal communication of healthcare professionals, such as doctors, physicians, nurses, and other medical practitioners, into written documents that become part of the patient's medical record. These transcribed documents are typically included in electronic health records (EHR) or other medical record-keeping systems(David et al., 2009).

Medical transcription encompasses a broad spectrum of critical information, encompassing patient histories, physical exam findings, diagnostic outcomes, treatment plans, prescribed medications, and pertinent medical particulars. This practice holds pivotal significance in documenting patient care, fostering interprofessional communication, and ensuring meticulous record-keeping for both legal and clinical purposes. Accomplished by trained medical transcriptionists well-versed in medical jargon and procedures, the process involves transcribing recorded dictations using specialised software. Profound familiarity with medical terminology, anatomy, physiology, and procedural nuances is imperative to guarantee the accuracy, completeness, and coherence of these transcriptions. Despite the emergence of electronic health record systems and technological advancements, the transition from manual to automated methods, the precise and lucid nature of medical transcriptions remains indispensable for upholding comprehensive patient records and facilitating effective communication within the healthcare sector(Anon, 2022).

The general structure of a medical transcription includes headings such as patient identification, date of service, history of present illness (HPI), review of systems (ROS), physical examination (PE), assessment and plan (A and P), and impression(Anon, 2022).

2.3. Need of medical transcription and its impact on healthcare sector:

Medical transcription holds a crucial position within the healthcare realm, playing an indispensable role across various dimensions. Patient medical records have become essential components in healthcare, aiding specialisation and coordination. Originating to cater to expanding personnel involved in patient care and billing, medical records have grown in significance with the complexity of healthcare.

They're vital for coordinated actions, ensuring accurate compensation, and enabling effective communication. Acting as "boundary objects," medical records transcend organisational boundaries, accessible to medical practitioners, billing professionals, researchers, and more. The accuracy and detail they embody are paramount given their diverse user base(David et al., 2009).

The transition from paper-based records to electronic health records has amplified their accessibility and storage efficiency. Amid this evolution, medical transcriptionists (MTs) play a pivotal role in transforming verbal interactions into written records, maintaining precision and conformity to medical standards(David et al., 2009).

A significant technological advancement is the introduction of Speech Recognition Technologies (SRTs), which transcribe spoken content into text. However, despite the benefits, SRTs are not infallible. MTs remain essential in editing and validating SRT-generated transcripts to ensure data integrity and quality(David et al., 2009).

The study delved into the practical advantages of leveraging medical transcription data to generate real-world evidence within the US healthcare system, focusing on the conversion of spoken medical notes into written records. Examining 19 records of adults with systemic lupus erythematosus (SLE) undergoing new belimumab treatment, the investigation unveiled valuable insights into treatment initiation, doctor-patient communication encompassing risk discussions and patient-driven research sharing. Notably, it exposed the influence of insurance and financial factors on treatment choices(Narayanan, 2013). Significant revelations encompassed patient experiences, with 42 percentage encountering treatment-related issues, a similar proportion discontinuing and later restarting, and almost half showcasing improvements such as pain reduction and increased energy, as observed by physicians. Ultimately, this study underscores the invaluable utility of medical transcription data in comprehending treatment commencement, interactions, and real-world outcomes, enabling informed medical decisions grounded in authentic patient experiences(Narayanan, 2013).

In summary, medical transcription and accurate record creation remain pivotal in modern healthcare. The shift to electronic records, incorporation of SRTs, and the expertise of MTs collectively contribute to comprehensive medical documentation, proficient communication among healthcare professionals, and optimal patient care.

2.4. What is text mining?

Text mining involves extracting concealed, unknown, and valuable insights from textual data. This emerging field aims to derive meaningful information from natural language, scrutinising text to extract details.

Text mining parallels data mining, but for unstructured or semi-structured datasets like emails, making it versatile. The process includes organising input text, incorporating features, and storing structured data. Patterns within are extracted, leading to evaluation(Kumar, 2013).

2.5. Process of text mining:

Pre-processing tasks are conducted to detect and eliminate irregularities. The cleansing process ensures the essential content of available text remains intact, involving the removal of stop words, stemming (root identification), and data indexing(Kumar, 2013).

Subsequent processing and management steps involve auditing and further refining the dataset using automated techniques. Pattern analysis is executed through a Management Information System (MIS).

The information processed in the aforementioned steps serves the purpose of extracting valuable and pertinent insights for timely decision-making and trend analysis. Extracting valuable information from a diverse document corpus is a demanding endeavour. Selecting suitable techniques to mine text reduces the time and effort required to uncover relevant patterns for analysis and decision-making(Talib et al., 2016).

Tokenization, entity extraction and vectorisation are the major text mining techniques used in the project.

2.5.1. Entity Extraction:

Entity recognition is a crucial task within Natural Language Processing (NLP) that involves identifying specific entities or elements, such as names of drugs, diseases, symptoms, route of administration, dosage forms, and more, within a given text(Ramachandran and Arutchelvan, 2021).

The significance of entity recognition lies in its ability to transform vast amounts of unstructured textual information into structured data(Ramachandran and Arutchelvan, 2021).

This enables researchers, healthcare professionals, and report generators to efficiently access, organise, and analyse valuable insights that are concealed within these documents.

Named Entity Recognition (NER) involves the application of predetermined rules to assign specific tags to words, corresponding to named entities representing individuals, locations, and associations. The primary objective of NER is to detect and classify both nominal and

numerical information within documents, categorizing words as individuals, organizations, or dates(Divakaran, 2019).

In the realm of linguistics, Named Entity Recognition refers to the automated process of scanning unstructured text to identify "entities." This aims to standardize terms and classify them into various categories like individual names, organizations (such as corporations, government bodies, committees), places (including cities, countries, rivers), or expressions indicating dates and times(Mansouri, Affendey and Mamat, 2008).

BioNER stands for Biomedical Named Entity Recognition and serves as the initial phase in the process of extracting connections between biological entities, which hold significance in medical investigation, such as gene-disease or disease-drug relationships. The significance of BioNER is growing within the field of biomedical research. It serves as the foundational step in identifying relationships between biological elements crucial to medical studies. For instance, gene-disease or disease-drug interactions(Perera, Dehmer and Emmert-Streib, 2020).

Biomedical text mining fundamentally involves three steps:

1) detecting biomedical entities like genes or diseases in text through named entity recognition(Rosário-Ferreira et al., 2021).

2) linking those entities to established databases by normalizing their names(Rosário-Ferreira et al., 2021).

3) extracting relationships between the identified concepts using relation extraction techniques(Rosário-Ferreira et al., 2021).

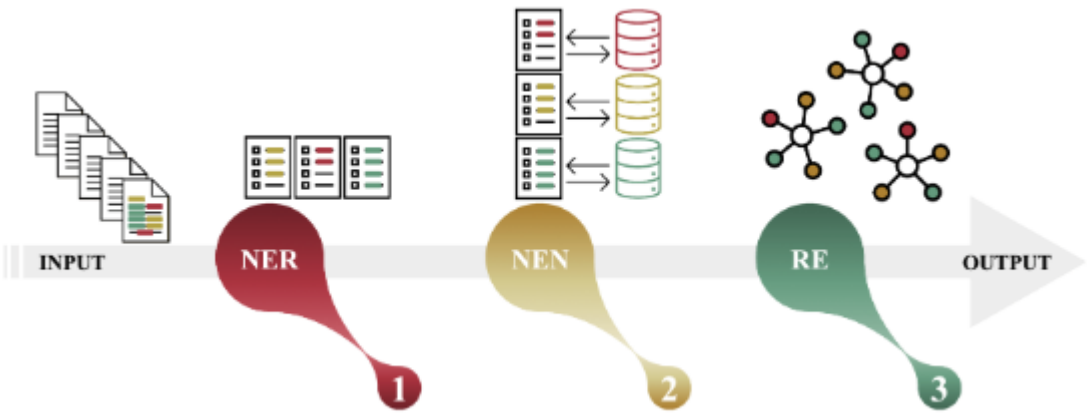

Figure 1: NER workflow

In the realm of linguistics, Named Entity Recognition refers to the automated process of scanning unstructured text to identify "entities." This aims to standardize terms and classify them into various categories like individual names, organizations, places, or expressions indicating dates and times(Mansouri, Affendey and Mamat, 2008).

scispaCy stands out as an optimal approach for pre-processing textual data, effectively priming it for deep learning applications. This freely accessible software library is purpose-built to facilitate advanced Natural Language Processing (NLP) tasks. Its core functionality revolves around delivering advanced processing capabilities tailored for textual data in the realms of biomedical, scientific, and clinical literature(Divakaran, 2019).

A scispaCy model designed for biomedical entity extraction is employed in this project to extract medical entities from medical transcription texts.

2.5.2. Tokenization:

Tokenization is a method used to break down a text into smaller units known as tokens. These tokens can incorporate words, characters, or sub words. Thus, tokenization can be categorised into three primary types: word, character, and sub word (n-gram characters) tokenization.

Word Tokenization is the most commonly used tokenization algorithm. It splits a piece of text into individual words based on a certain delimiter.

For instance, let's consider the sentence: "why so serious"

The prevalent approach for creating tokens relies on spaces. If we use space as a divider, the sentence tokenization yields 3 tokens: why, so and serious. Since each token corresponds to a word, this exemplifies the process of word tokenization(Pai, 2020).

However, a limitation of tokenization arises when attempting to tokenize documents that lack whitespace, special characters, or other markers. The demand for tokenization originates from the inherent nature of textual data, initially existing as a string of characters. Subsequent text analysis processes heavily rely on the presence of words within the dataset(Mohan, 2016).

This project leverages the Natural Language Toolkit (NLTK) for tokenization. NLTK is a well-regarded Python Natural Language Processing Toolkit, serving as a pivotal platform for developing Python programs that engage with human language data. Notably, NLTK provides intuitive interfaces for over 50 corpora, encompassing structured text sets, as well as lexical resources like WordNet(Mohan, 2016).

Furthermore, NLTK encompasses an array of text processing libraries tailored for various tasks, including classification, tokenization, stemming, tagging, parsing, and semantic reasoning. In this project, NLTK is specifically employed for tokenization purposes.

2.5.3. Vectorisation:

Vectorization is a process that converts text data into a numerical format, making it compatible with machine learning algorithms designed to work with numerical information. This conversion bridges the gap between textual and numerical domains by representing text as numerical vectors(Chandrayan, 2021).

Techniques like Bag of Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF), and Word Embeddings (such as Word2Vec and GloVe) offer distinct approaches to this process, catering to various NLP tasks and data types(Chen, 2020).

Ultimately, vectorization empowers machine learning models to decode and analyse text, uncovering patterns and information within extensive textual datasets.

In this project TF-IDF vectorisation is used for vectorisation,

TF-IDF, which stands for term frequency-inverse document frequency, is an algorithm designed to assess the significance of keywords within a corpus of documents. It plays a crucial role in determining the relevance of specific keywords to individual documents, enabling the identification and categorization of documents based on anticipated keywords(Chandrayan, 2021).

Term frequency evaluates the frequency of a term's occurrence within a text, offering insight into its prominence. In contrast, inverse document frequency measures the rarity of a term across the entire corpus, highlighting its uniqueness. By assigning higher weights to terms that are both prevalent within a document and distinctive across the corpus, TF-IDF enhances the ability to differentiate documents based on their content(Chandrayan, 2021).

TF(t) = (Number of times term t appears in a document) / (Total number of terms in the document)(LessenText, 2023)

IDF(t) = loge(Total number of documents) / (Number of documents with term t in it)(LessenText, 2023)

For example, to analyse the importance of terms within the collection of patient medical records, Text 1, text 2 and text 3.

Text 1: "Ibuprofen is prescribed to manage the patient's pain. The patient reports occasional headaches."

Text 2: "The patient's blood pressure is within normal ranges. The patient's diet and exercise routine are discussed."

Text 3: "The patient's symptoms include fever and cough. A chest X-ray is ordered for further diagnosis."

The calculated TF-IDF scores for the term "patient" within these documents:

Term Frequency (TF): These measures how frequently the term "patient" appears in each document, divided by the total number of terms in that document.

Text 1: "patient" appears 2 times, total terms = 15. TF = 2/15 approximately equals 0.133.

Text 2: "patient" appears 2 times, total terms = 14. TF = 2/14 approximately equals 0.143.

Text 3: "patient" appears 2 times, total terms = 13. TF = 2/13 approximately equals 0.154.

Inverse Document Frequency (IDF): This measures the rarity of the term "patient" across the entire collection of documents. It's calculated as the logarithm of the ratio between the total number of documents and the number of documents containing the term "patient."

Total Texts = 3

Number of Texts containing "patient" = 3

IDF = log (3 / 3) = 0

TF-IDF Score: The final TF-IDF score is the product of the TF and IDF values for each document.

Document 1: TF-IDF = TF * IDF = 0.133 * 0 = 0

Document 2: TF-IDF = TF * IDF = 0.143 * 0 = 0

Document 3: TF-IDF = TF * IDF = 0.154 * 0 = 0

In this example, the TF-IDF scores for the term "patient" are all 0 because the IDF value for the term is 0. This is because the term "patient" appears in every document, making it a common term and causing the IDF to be 0.

2.5.4. Dimensionality reduction:

Dimensionality refers to the number of attributes that define each data point. High-dimensional data, common in fields like text mining, can be intricate and computationally demanding. Techniques like Principal Component Analysis (PCA) are used to reduce dimensions while retaining relevant information, simplifying analysis, and enhancing efficiency.

Principal Component Analysis is a method for deriving new features through linear combinations of original features. It transforms instances from a d-dimensional space to a lower-dimensional subspace with k dimensions (k < d). The resulting k dimensions are known as Principal Components (PCs), capturing maximum variance while minimising overlap with previously considered components(Keerthi Vasan and Surendiran, 2016).

PCA has many advantages such as removing correlated features that undermine the statistical significance of an independent variable, improving algorithm performance, which can be significantly degraded if too many features are present in models, and speeding up analyses(Abdel-Fattah et al., 2021).

PCA is a statistical method that employs an orthogonal transformation to convert a collection of observations, which could involve correlated variables, into a new set of linearly uncorrelated variables called principal components. This transformation is structured to achieve two key objectives: the first principal component captures the maximum variance possible, effectively accounting for the primary variability in the data, while the successive components maximise variance while also being orthogonal, or uncorrelated, with the preceding components. In essence, PCA simplifies complex data by creating a new set of variables that efficiently encapsulate the significant patterns within the original data, aiding in dimensionality reduction and revealing crucial relationships among variables(Shlens, 2014).

2.6. Text mining in healthcare sector:

Text mining is vital in healthcare due to data volume from IT services. Healthcare's importance grows with disease surge, yielding data in hospitals, clinics, insurance, and more. Healthcare data expands with tech like wearables, genomics, and drug discovery, forming Healthcare Big Data. EMRs hold info from institutions and clinics, structured (demographics, labs) and unstructured (narratives, images). EMRs aid patient care, research, development, and emergencies.

Given the diverse nature of medical records, extracting valuable insights is essential, making text mining a pivotal tool. Text mining techniques unveil connections between symptoms, diseases, treatments, and diagnostic tests, supporting early disease detection, refining treatments, and curbing expenses(Mathuna and Shanthi, 2021).

Some of the challenges faced by researchers in this area include data heterogeneity, the presence of domain-specific language, large label space, and the need for effective validation and evaluation methods(Gonzalez et al., 2012).

Analysing electronic medical records (EMR), medical claims, prescriptions, and images through text mining can reveal patterns in patient care. This helps pinpoint care gaps like missed diagnoses or treatment non-compliance, enabling healthcare providers to enhance patient care, reduce gaps, and prevent adverse events(Green, Cary and Nc 27518 919.459.4600, n.d.).

The case study conducted to figure out how genomic testing influences treatment decisions and trends in mental health disorders utilised Real Health Data, a comprehensive medical transcription database covering the US. This database contains detailed narrative medical records that offer insights into patient conditions and interactions with physicians. The researchers employed natural language processing techniques like text mining to search for instances of genetic testing and mental health disorders like depression, schizophrenia, anxiety, and ADHD within the database(Mehta et al., 2018).

The research concluded that medical transcription data, when combined with other types of health data like claims, electronic medical records (EMR), and genomic data, could offer valuable insights into the field of health economics and outcomes research (HEOR). These insights could guide treatment decisions for psychiatric disorders and contribute to the broader movement of treating the individual rather than just the illness(Mehta et al., 2018).

2.7. Class imbalance in medical dataset:

In the realm of healthcare data, class imbalance arises when one particular class, typically representing a minority group, is significantly underrepresented compared to another class. This disparity introduces complexities for machine learning and classification algorithms, as their performance might falter when handling instances from the minority class(Rahman and Davis, 2013).

To confront this challenge, sampling techniques are harnessed, encompassing either the reduction of instances in the majority class (under-sampling) or the augmentation of instances in the minority class (over-sampling)(Rahman and Davis, 2013).

Approaches such as SMOTE (Synthetic Minority Over-sampling Technique) in over-sampling generate synthetic instances of the minority class to attain dataset equilibrium, whereas under-sampling entails the random selection of a subset from the majority class. Both approaches carry their own merits and drawbacks, influencing factors such as the size of the training set, computational time, and the potential for overfitting(Rahman and Davis, 2013).

Within this project, the SMOTE technique is strategically employed to mitigate the class imbalance present within the transcription dataset.

2.8. Use of machine learning to classify medical data

The machine learning approach emphasises using sophisticated computational methods to recognize data, while the utilisation of natural language processing techniques enables the handling and examination of textual information composed in human languages(Pagad et al., 2022). Logistic regression, Decision tree, Support vector machine, Random Forest, K Nearest Neighbour and Naïve bayes are the most commonly used machine learning algorithms(Gong, 2022).

Two machine learning algorithms Logistic regression and random forest are mainly used on the data and the accuracy of the prediction is calculated and compared.

Logistic regression is a statistical model used for classification and predictive analytics. It estimates the probability of an event occurring based on a given dataset of independent variables. The dependent variable is bounded between 0 and 1 since the outcome is a probability(Anon, n.d.). In logistic regression, a logit transformation is applied on the odds—that is, the probability of success divided by the probability of failure. This is also commonly known as the log odds, or the natural logarithm of odds.

A random forest is an algorithm that involves combining multiple tree predictors. Each tree's construction is influenced by values from a random vector, drawn independently with the same distribution across all trees(Breiman, 2001). In the fields of biology and medicine, random forests have effectively addressed various challenges. These include forecasting the reaction of drugs in cancer cell lines(Riddick et al., 2011) and detecting DNA-binding proteins. Random forests have also recognized handwritten digits with high accuracy(Denisko and Hoffman, 2018).

2.9. Gap of Knowledge:

Limited Application to Specific Medical Specialties: Certain medical specialties or domains within medical transcription have received extensive study, while others remain underrepresented. The majority of studies focus on general medical texts, with limited data available for specialised fields.

Challenges in Handling Noisy Data: Medical transcription data is inherently noisy due to abbreviations and the use of repeated words. There is limited research addressing these challenges in the context of text mining and machine learning.

Limited Incorporation of Domain-Specific Knowledge: When it comes to effectively integrating domain-specific medical knowledge, there is a need to explore how to leverage medical ontologies, terminologies, or expert knowledge to enhance classification accuracy.

Insufficient Exploration of Feature Engineering Techniques: The effectiveness of text mining and machine learning models often relies on well-designed features. There's a lack of exploration into innovative feature engineering techniques tailored to medical transcription data.

Sparse Attention to Imbalanced Data Issues: Class imbalance is common in medical datasets, where certain medical conditions might be rare compared to others.

Real-World Clinical Impact: While many studies might focus on achieving high classification accuracy, there is a gap in understanding how these methods translate to real-world clinical settings. Investigating the actual impact of classification on medical decision-making and patient outcomes is needed.

2.10. Conclusion:

In conclusion, this literature review navigated the terrain of medical transcription, text mining, and technology's role in healthcare. It emphasised the pivotal role of accurate medical records in modern healthcare. Exploring text mining techniques highlighted their power to get insights from medical text, aiding classification and decision-making. The challenge of class imbalance within medical data was addressed, and the integration of machine learning for medical text classification was analysed. Balancing technology with domain knowledge is essential for optimising patient care and healthcare practices.

3. METHODOLOGY

3.1. Problem definition

Medical transcriptions contain vital patient information, but their manual analysis is resource-intensive and susceptible to errors. This project uses the potential of automation by employing text mining methods to extract essential medical entities and patterns from transcriptions. It also aims to develop predictive models capable of associating each transcription with a specific medical specialty.

The central aim of this project is to apply advanced text mining techniques to medical transcriptions and utilise the derived insights to predict the corresponding medical specialty. Additionally, the project aims to assess the accuracy of various machine learning algorithms and compare their performance to determine the optimal machine learning approach. Furthermore, the project aims to explore the feasibility and potential of classifying medical transcriptions through the integration of text mining and machine learning.

3.2. Data collection

A dataset containing several columns relating to medical transcriptions and their corresponding medical specialties has been acquired from Kaggle, a popular platform for data science resources. This dataset consists of 4,998 rows and 5 columns. The columns present in the dataset include 'description,' 'medical specialty,' 'sample name,' 'transcription,' and 'keywords.' The primary focus of our analysis involves the 'medical specialty' and 'transcription' columns.

The dataset link has been downloaded from:

Medical Transcriptions | Kaggle

The data within this dataset originates from 'MTSamples,' a website that provides access to an extensive collection of transcribed medical reports.

The website aims to provide valuable materials for individuals who are either studying or actively engaging in medical transcription, supporting both learning and practical application(Anon, n.d.).

3.3. Tools and techniques

The code execution was conducted within the Google Colaboratory environment, which provides a cloud-based platform for running code, particularly well-suited for data science tasks.

The programming language employed for writing and executing the code is Python. Python is a widely-used, versatile, and beginner-friendly programming language, making it a popular choice for various applications, including data analysis, machine learning, and scientific

computing. Its extensive libraries and frameworks, such as scikit-learn and pandas, facilitate seamless data manipulation, model development, and evaluation.

spacy: A natural language processing library that provides pre-trained models and tools for processing and analysing text data, including part-of-speech tagging, named entity recognition (NER), and more.

en_ner_bionlp13cg_md: A specific spaCy model trained for biomedical entity recognition, aiding in accurately identifying biomedical terms and concepts in text.

pandas: A versatile data manipulation library that provides data structures and functions for efficiently working with structured data, such as data frames, tables, and various data analysis operations.

numpy: A fundamental library for numerical computations in Python, providing support for arrays, matrices, and mathematical functions.

matplotlib: A popular visualization library that allows for creating static, interactive, and animated visualizations in Python, enabling data presentation in graphical formats.

seaborn: Built on top of matplotlib, seaborn is a statistical data visualization library that offers a high-level interface for creating informative and attractive statistical graphics.

string: A built-in Python module that provides a collection of string constants and helper functions for string manipulation and formatting.

re: The regular expression module in Python, which offers support for pattern matching and manipulation of strings based on regular expressions.

sklearn: Scikit-learn, a machine learning library, provides various tools for data pre-processing, modelling, evaluation, and more.

nltk: The Natural Language Toolkit is a library for natural language processing, offering functionalities for tokenization, stemming, part-of-speech tagging, and other language-related tasks.

imblearn: A library specifically designed to address class imbalance issues in machine learning, offering techniques like SMOTE (Synthetic Minority Over-sampling Technique).

CountVectorizer: A scikit-learn class for converting text data into a matrix of token counts, a common step in text data pre-processing for machine learning.

cosine_similarity: A function from scikit-learn to compute the cosine similarity between vectors, often used in measuring similarity between text documents.

tabulate: A Python library for formatting and printing tabular data, often used to create tables for data presentation.

LabelEncoder: A utility from scikit-learn for converting categorical labels into numerical values, typically used to prepare target variables for machine learning models.

gc: The garbage collection module in Python, used for controlling the behaviour of the garbage collector and managing memory.

3.4. Load dataset

The dataset is loaded as pandas data frame with name mtdata_df using pandas library in python.

Pandas is a Python library that provides rich data structures and tools for working with structured data sets common to statistics, finance, social sciences, and many other fields . It aims to be the foundational layer for the future of statistical computing in Python (pandas). It also complements the existing scientific Python stack, such as NumPy, SciPy, and matplotlib(Mckinney, 2011).

3.5. Understanding the data

The loaded pandas data frame is analysed by understanding the number of rows and columns in total and found that there are 4998 rows and 5 columns.

description	medical_specialty	sample_name	transcription	keywords
A 23-year-old white female presents with comp...	Allergy / Immunology	Allergic Rhinitis	SUBJECTIVE:, This 23-year-old white female pr...	allergy / immunology, allergic rhinitis, aller...
Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult - 2	PAST MEDICAL HISTORY:, He has difficulty climb...	bariatrics, laparoscopic gastric bypass, weigh...
Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult - 1	HISTORY OF PRESENT ILLNESS: , I have seen ABC ...	bariatrics, laparoscopic gastric bypass, heart...
2-D M-Mode. Doppler.	Cardiovascular / Pulmonary	2-D Echocardiogram - 1	2-D M-MODE: , ,1. Left atrial enlargement wit...	cardiovascular / pulmonary, 2-d m-mode, dopple...
2-D Echocardiogram	Cardiovascular / Pulmonary	2-D Echocardiogram - 2	1. The left ventricular cavity size and wall ...	cardiovascular / pulmonary, 2-d, doppler, echo...

Figure 2: Loaded Dataset

Identified the target variable as ‘medical_specialty’ which has many classes.

Found that there are 39 classes of medical specialities in the column using the unique() method.

```
([' Allergy / Immunology', ' Bariatrics',
  ' Cardiovascular / Pulmonary', ' Dentistry', ' Urology',
  ' General Medicine', ' Surgery', ' Speech - Language',
  ' SOAP / Chart / Progress Notes', ' Sleep Medicine',
  ' Rheumatology', ' Radiology', ' Psychiatry / Psychology',
  ' Podiatry', ' Physical Medicine - Rehab',
  ' Pediatrics - Neonatal', ' Pain Management', ' Orthopedic',
  ' Ophthalmology', ' Office Notes', ' Obstetrics / Gynecology',
  ' Neurosurgery', ' Neurology', ' Nephrology', ' Letters',
  ' Lab Medicine - Pathology', ' IME-QME-Work Comp etc.',
  ' Hospice - Palliative Care', ' Hematology - Oncology',
  ' Gastroenterology', ' ENT - Otolaryngology', ' Endocrinology',
  ' Emergency Room Reports', ' Discharge Summary',
  ' Diets and Nutritions', ' Dermatology',
  ' Cosmetic / Plastic Surgery', ' Consult - History and Phy.',
  ' Chiropractic'], dtype=object)
```

Figure 3: list of medical specialties

It is noted that certain categories are irrelevant and some categories can be made into one class. Such filtration processes are later done in the filtration step.

A bar chart was created to display the occurrence count of each medical specialty class. It was observed that the "surgery" class exhibited the highest frequency among all specialties,

followed by "Orthopaedic," "cardiovascular/pulmonary," "radiology," "consult history and phy.," "gastroenterology," "neurology," "general medicine," and the remaining classes, respectively, in descending order.

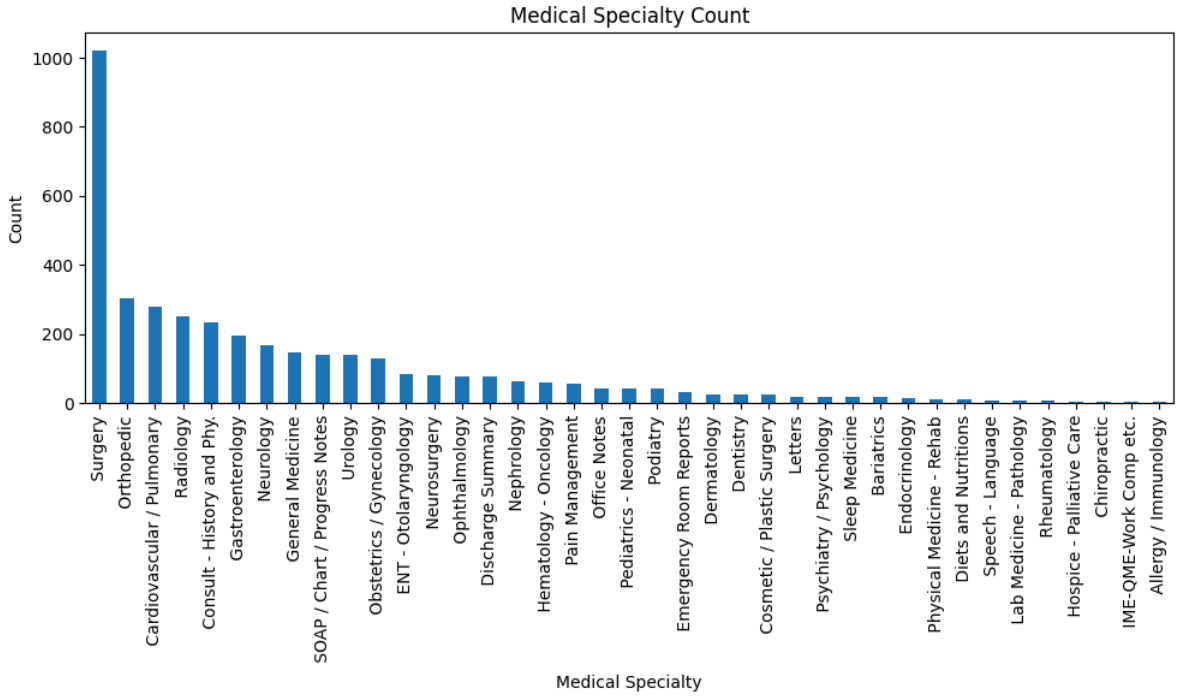

Figure 4: Bar chart showing count of medical specialities

The bar chart clearly indicates a substantial data imbalance among various classes. Several categories have notably low occurrences within the data frame. Particularly, the "Allergy/Immunology" specialty exhibits the lowest frequency. This could potentially be attributed to the underutilization of transcriptions within this medical field.

It's worth highlighting those certain significant specialties, such as endocrinology, have a notably small dataset representation. This observation raises questions about the comprehensive integration of medical transcription practices within this domain, suggesting a potential lack of recognition of its relevance.

Upon initial inspection of the unique classes, it becomes evident that certain classes hold little relevance. For instance, classes such as "Speech – Language," "SOAP / Chart / Progress Notes," "Sleep Medicine," "Physical Medicine – Rehab," "Pain Management," "Office Notes," "Letters," "Lab Medicine - Pathology," "IME-QME-Work," "Hospice - Palliative Care," "Emergency Room Reports," "Discharge Summary," "Diets and Nutrition," "Cosmetic / Plastic Surgery," and "Consult - History and Phy." are identified as irrelevant for the project's objectives. Consequently, these classes were later excluded during the filtration step. While the mentioned classes possess significance, their inclusion introduces unwanted noise into the dataset that could hinder the project's purpose. Therefore, they will be removed to enhance the dataset's relevance and focus on the project's specific goals.

3.6. Data cleaning and preparation

cleaning and pre-processing are essential steps to prepare raw data for further analysis. These steps involve transforming the data into a usable format by removing irrelevant or noisy information, correcting errors, and enhancing the quality of the data. The goal is to ensure that the data is accurate, consistent, and ready for analysis, thus improving the reliability of insights derived from it.

It is a step-by-step procedure that involves various stages, including defining data transformation and mapping rules, performing transformations, verifying the changes, and conducting a retrospective data analysis(National University of Computer and Emerging Sciences, Faisalabad, 38000, Pakistan et al., 2017).

There are 5 columns in which only medical specialty and transcription columns are needed, the remaining columns are deleted. Similarly, it has been found that the transcription column contains 33 null values. These rows containing null values have been deleted.

3.6.1. Cleaning:

The transcription column has text as shown in figure 5,

```
'SUBJECTIVE:,  This 23-year-old white female presents with complaint of allergies.  She used to have al
lergies when she lived in Seattle but she thinks they are worse here.  In the past, she has tried Clari
tin, and Zyrtec.  Both worked for short time but then seemed to lose effectiveness.  She has used Alleg
ra also.  She used that last summer and she began using it again two weeks ago.  It does not appear to
be working very well.  She has used over-the-counter sprays but no prescription nasal sprays.  She does
have asthma but doest not require daily medication for this and does not think it is flaring up.,MEDICA
TIONS: , Her only medication currently is Ortho Tri-Cyclen and the Allegra.,ALLERGIES: , She has no kno
wn medicine allergies.,OBJECTIVE:,Vitals:  Weight was 130 pounds and blood pressure 124/78.,HEENT:  Her
throat was mildly erythematous without exudate.  Nasal mucosa was erythematous and swollen.  Only clear
drainage was seen.  TMs were clear.,Neck:  Supple without adenopathy.,Lungs:  Clear.,ASSESSMENT:,  Alle
rgic rhinitis.,PLAN:,1.  She will try Zyrtec instead of Allegra again.  Another option will be to use l
oratadine.  She does not think she has prescription coverage so that might be cheaper.,2.  Samples of N
asonex two sprays in each nostril given for three weeks.  A prescription was written as well.<'
```

Figure 5: Transcription as in the dataset before cleaning

The data frame is analysed and found that the 'transcription' column, which contains medical transcriptions that are highly noisy. These transcriptions include various special characters that contribute to the noise and complexity of the text.

The text is converted to lowercase and also is converted to string format. String format is a common way of representing unstructured text data in programming languages and databases. String format allows text to be split into tokens, such as words or phrases, that can be used as inputs for machine learning algorithms. String format enables text to be cleaned and normalised, such as by removing punctuation, lowercasing, stemming, lemmatizing, etc., to reduce noise and variability in the data.

These characters are removed in the cleaning step by replacing every instance of characters found within the 'transcription' column with spaces. This step is fundamentally aimed at cleaning the textual data by effectively eliminating undesired characters. This step will enhance the cleanliness and manageability of the dataset, rendering it more suitable for subsequent analysis.

```
'subjective  this yearold white female presents with complaint of allergies  she used to have allergies
when she lived in seattle but she thinks they are worse here  in the past she has tried claritin and zy
rtec  both worked for short time but then seemed to lose effectiveness  she has used allegra also  she
used that last summer and she began using it again two weeks ago  it does not appear to be working very
well  she has used overthecounter sprays but no prescription nasal sprays  she does have asthma but doe
st not require daily medication for this and does not think it is flaring upmedications  her only medic
ation currently is ortho tricyclen and the allegraallergies  she has no known medicine allergiesobjecti
vevitals  weight was  pounds and blood pressure heent  her throat was mildly erythematous without exuda
te  nasal mucosa was erythematous and swollen  only clear drainage was seen  tms were clearneck  supple
without adenopathylungs  clearassessment  allergic rhinitisplan  she wi… '
```

Figure 6: Cleaned text

3.6.2. Lemmatization:

The cleaned transcription is subsequently subjected to lemmatization, a process that converts the text data into its normalised form. This normalised form represents the root or base form of each word present in the transcription(Plisson, Lavrac and Mladenic, n.d.).

Here we use NLTK (Natural Language Toolkit) which is a python library. NLTK offers the WordNetLemmatizer class, which serves as a concise interface built upon the WordNetCorpus. This class utilises the Morphy() function within the WordNetCorpusReader to identify the base word or lemma(Khyani and B S, 2021).

For example, the lemma of word “Dogs” is “Dog”, and the lemma of “walking” is “walk”(Anon, 2018).

Figure 7 shows the lemmatized text,

```
'subjective this yearold white female present with complaint of allergy she used to have allergy when s
he lived in seattle but she think they are worse here in the past she ha tried claritin and zyrtec both
worked for short time but then seemed to lose effectiveness she ha used allegra also she used that last
summer and she began using it again two week ago it doe not appear to be working very well she ha used
overthecounter spray but no prescription nasal spray she doe have asthma but doest not require daily me
dication for this and doe not think it is flaring upmedications her only medication currently is ortho
tricyclen and the allegraallergies she ha no known medicine allergiesobjectivevitals weight wa pound an
d blood pressure heent her throat wa mildly erythematous without exudate nasal mucosa wa erythematous a
nd swollen only clear drainage wa seen tm were clearneck supple without adenopathylungs clearassessment
allergic rhinitisplan she will try zyrtec instead of allegra again another option will be to use lorata
dine she doe not think she ha prescription coverage so that might be cheaper sample of nasonex two spra
y in each nostril given for three week a prescription wa written a well'
```

Figure 7: Lemmatized text

3.6.3. Entity extraction

The lemmatized text undergoes processing using the "en_ner_bionlp13cg_md" scispaCy model. This model specialises in recognizing biomedical entities. The goal is to extract medical terms from transcriptions, enhancing understanding and analysis. To streamline this process, a dedicated function is crafted.

This function takes the lemmatized text from the transcription column as input and executes the following steps: applying the scispaCy model, identifying medical entities like diseases, drugs, and genes, and finally, extracting these entities.

The entire procedure facilitates automated extraction of medical terminologies, pivotal for medical research, information retrieval, and domain-specific applications(\cite{allenai2020labelissue}).

After applying scispaCy model, it will extract only the medical terms and thus the text becomes like,

claritin nasal blood erythematous exudate nasal mucosa wa erythematous loratadine nostril

Here we can see that the model successfully extracted the medical terms from the lemmatized text.

3.6.4. Exploratory Data Analysis

A detailed exploratory data analysis is conducted on the pre-processed data frame.

Exploratory Data Analysis (EDA) is an approach to analysing data that focuses on understanding its inherent patterns and characteristics without being constrained by specific models or assumptions. Unlike traditional statistical methods, EDA does not require an explicit underlying population model. Instead, it treats data as a list of values and employs various techniques, such as graphical visualisation and summary statistics, to gain insights from multiple perspectives(Morgenthaler, 2009).

The average similarity is calculated from the table as shown in Figure 8 for each specialty, providing insights into the impact of certain categories on the overall noise analysing the average cosine similarity between medical specialties by calculating word frequency vectors from transcriptions using scikit-learn's CountVectorizer and cosine_similarity functions, and presents the results in a formatted table using the tabulate library.

Medical Specialty	Average Similarity	Number of Rows
Consult - History and Phy.	0.7818107537666906	234
Surgery	0.7676640690250817	1021
Emergency Room Reports	0.7672190673434834	31
SOAP / Chart / Progress Notes	0.7652304955312509	142
General Medicine	0.7630921812565118	146
Discharge Summary	0.7532054495298153	77
Hematology - Oncology	0.7425322991796715	62
Pediatrics - Neonatal	0.7413558043503853	42
Pain Management	0.7282342708411724	58
Psychiatry / Psychology	0.7253515114584176	19
IME-QME-Work Comp etc.	0.721564885987878	4
Speech - Language	0.7131684235051556	8
Orthopedic	0.7101013455564507	303
Hospice - Palliative Care	0.7015084498639677	5
Sleep Medicine	0.701446359320613	18
Dermatology	0.692750908423141	25
Urology	0.6807338754010804	140
Gastroenterology	0.6753480400525429	195
Letters	0.673480623152993	20
Office Notes	0.668482060303621	44
Neurology	0.6608903587011274	168
Chiropractic	0.6538159883590103	4
Bariatrics	0.6443326317405412	18
Neurosurgery	0.6396523115062445	81
Obstetrics / Gynecology	0.6393875485500313	130
Diets and Nutritions	0.6310122070479202	10
Dentistry	0.6280542692890324	25
Cardiovascular / Pulmonary	0.6251397346938729	280
Nephrology	0.623743452615337	63
Podiatry	0.6185976658500675	42
ENT - Otolaryngology	0.61712412602732	84
Radiology	0.6126828288975309	251
Cosmetic / Plastic Surgery	0.5930609940520819	25
Endocrinology	0.5681498859556734	15
Physical Medicine - Rehab	0.5015336747840091	11
Ophthalmology	0.4619319258994517	79
Rheumatology	0.3808345438868961	7
Allergy / Immunology	0.3547582070431094	3
Lab Medicine - Pathology	0.2166602685974677	8

Figure 8: Table showing average similarity of words in each speciality

In the table, the medical specialties are sorted in descending order based on their average similarity. It is evident that certain specialties have an average similarity of over 0.7, and a significant portion of these fall under irrelevant categories.

Some medical specialties exhibit relatively high average similarity values. For instance, "Consult - History and Phy.," "Surgery," and "Emergency Room Reports" have relatively higher average similarity scores. This indicates that transcriptions within these specialties share common medical terminologies and language patterns.

On the other hand, specialties like "Allergy / Immunology," "Rheumatology," and "Lab Medicine - Pathology" have notably lower average similarity scores. This suggests that transcriptions within these specialties display more diversity in language usage.

It's evident that the number of rows varies significantly across specialties. Some specialties, such as "Allergy / Immunology" and "Rheumatology," have very few rows, which might pose challenges for accurate predictions due to data imbalance.

Furthermore, specialties such as "Surgery" had the chance to have words associated with a wide range of medical scenarios, such as human anatomy, organs, and medical procedures. This widespread applicability could result in less distinctive language patterns, thereby

reducing the effectiveness of the subsequent analysis in accurately predicting medical specialties.

It also hints at potential clusters of specialties with similar language patterns. These clusters could be explored further to understand underlying relationships between specialties.

The medical specialties showing average similarity greater than 0.7 is deleted during filtration, and after that a heat map is generated to visualize the frequency of shared words among different medical specialties as shown in Figure 9.

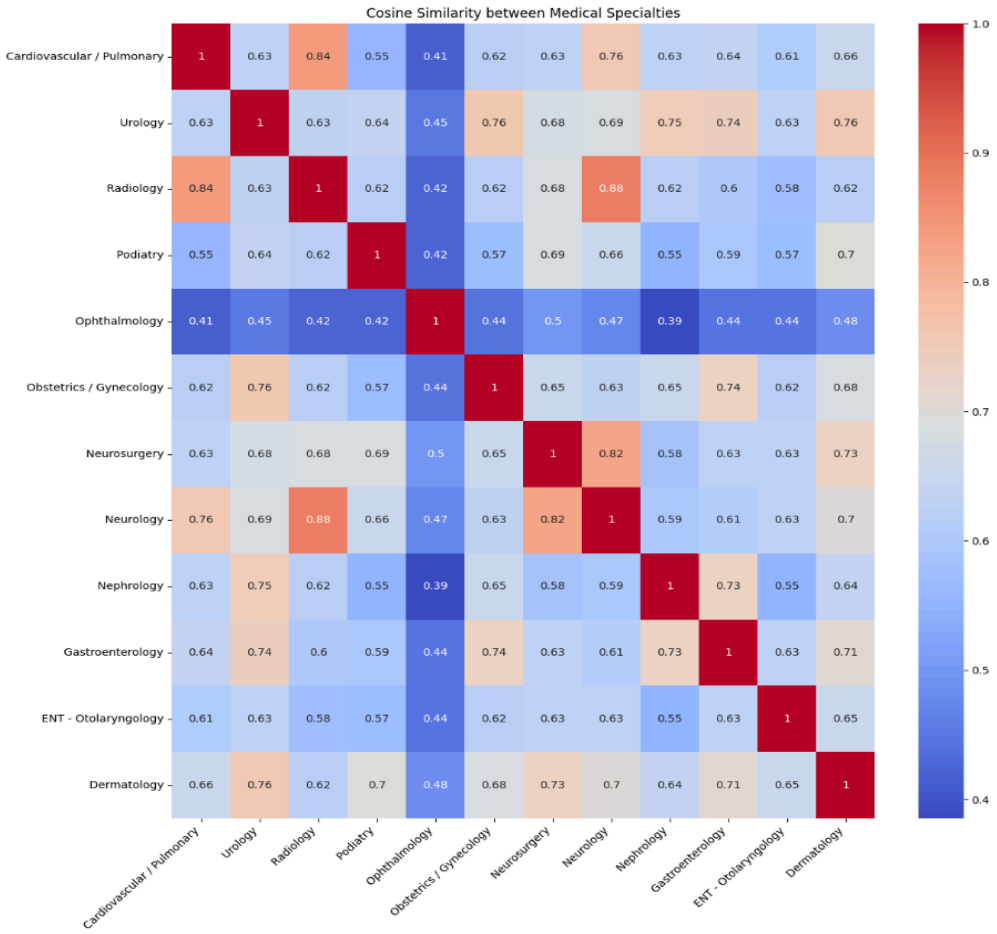

Figure 9: Heat map showing correlation of words between medical specialties

Upon analysing the heat map, it is evident that Neurology and Radiology exhibit a similarity score of 0.88. Consequently, both disciplines can be grouped under the same specialty, namely Neurology. Similarly, Radiology demonstrates a similarity score of 0.84 with Cardiovascular / Pulmonary. The results suggest that Radiology shares common terms with multiple specialties, indicating its potential linkage with various medical domains.

3.6.5. Filtration

The process of refining the dataset involved the exclusion of specific medical specialties for several reasons. Firstly, certain medical specialties exhibited an average word similarity greater than 0.7, indicating that the language used in their transcriptions was highly similar. To maintain the diversity of the dataset and prevent potential bias in subsequent analyses, these specialties were removed.

Additionally, some medical specialties were less relevant to the core objectives of the analysis. For example, "Surgery" and "Emergency Room Reports" were excluded from the dataset. These specialties were prone to contain language and terminology that were common across various medical contexts, leading to potential ambiguity in the predictive modelling phase.

By carefully eliminating these specialties, the dataset was refined to enhance the uniqueness of language patterns within each medical specialty. This strategic filtration ensured that the subsequent analyses and predictive modelling would be based on more distinct and relevant data, thereby improving the accuracy and interpretability of the outcomes.

By applying a little domain knowledge, we can do some filtration,

Neurosurgery and Neurology are interrelated and most data are interrelated so they both are combined as Neurology. The heat map also proves the same where the similarity is 0.82.

Similarly, Nephrology and Urology are also combined as Urology.

The analysis of the number of occurrences for each medical specialty revealed that several specialties have fewer than 50 occurrences. Due to the limited representation of these specialties in the dataset, their inclusion might lead to unreliable or biased results. Therefore, to ensure the robustness of the analysis, these specialties with fewer than 50 occurrences were excluded.

The remaining specialties are shown in Figure 10.

```
Cardiovascular / Pulmonary : 280
ENT - Otolaryngology : 84
Gastroenterology : 195
Neurology : 249
Obstetrics / Gynecology : 130
Ophthalmology : 79
Radiology : 251
Urology : 203
```

Figure 10: Remaining specialties after filtration

3.6.6. Tokenization

Tokenization is implemented to further analyse the remaining text. The text after entity extraction is subjected to tokenization. The word_tokenize function from the nltk.tokenize library is employed to tokenize the text.

word_tokenize function is designed to return a tokenized version of the provided text. NLTK's recommended word tokenizer, which consists of an enhanced TreebankWordTokenizer

combined with PunktSentenceTokenizer for the specific language, is employed in this process(cite{nltktokenize}).

The text before tokenization appears as follows,

ventricle left ventricular aortic valve mitral valve pulmonary valve pa systolic pressure mitral tricuspid aortic pulmonary

and after tokenization, it takes the following form.

'ventricle', 'left', 'ventricular', 'aortic', 'valve', 'mitral', 'valve', 'pulmonary', 'valve', 'pa', 'systolic', 'pressure', 'mitral', 'tricuspid', 'aortic', 'pulmonary'

The tokenized words are stored in a newly created column called "tokenized_words." The format of the tokenized words is an array.

3.6.7. part of speech tagging

The POS tags associated with each token in the tokenized words are determined using the POS tagging feature from NLTK. The averaged perceptron tagger is downloaded from NLTK and the resulting tags are stored in a new column called 'POSTags'.

The Averaged Perceptron Tagger is a sequence tagger based on an averaged perceptron model 1. It is the default tagger as of NLTK version 3.1 1(Anon, n.d.).

The tagger is used to assign part-of-speech tags to words in a text . The Averaged Perceptron Tagger is a good choice for text mining because it is fast, accurate, and easy to use(Spoustová et al., 2009).

By applying this function, it gives a new tuple. Each tuple contains two elements: the first element is a word/token, and the second element is the associated Part-of-Speech (POS) tag. The list as a whole is a collection of these tuples.

The result of the averaged perceptron tagger looks like this,

('left', 'VBN'), ('ventricular', 'JJ'), ('wall', 'NN'), ('wall', 'NN'), ('left', 'VBD'), ('ventricular', 'JJ'), ('left', 'VBD'), ('ventricular', 'JJ'), ('left', 'VBD'), ('ventricular', 'JJ'), ('left', 'VBD'), ('atrial', 'JJ'), ('left', 'JJ'), ('atrium', 'NN'), ('atrium', 'NN'), ('ventricle', 'NN'), ('aortic', 'JJ'), ('root', 'NN'), ('aortic', 'JJ'), ('valve', 'NN'), ('aortic', 'JJ'), ('valve', 'NN'), ('aortic', 'JJ'), ('valve', 'NN'), ('area', 'NN'), ('mitral', 'JJ'), ('annular', 'JJ'), ('leaflet', 'NN'), ('mitral', 'JJ'), ('valve', 'NN'), ('mitral', 'JJ'), ('tricuspid', 'JJ'), ('valve', 'NN'), ('tricuspid', 'JJ'), ('pulmonary', 'JJ'), ('artery', 'NN'), ('pulmonary', 'JJ'), ('artery', 'NN'), ('pulmonary', 'JJ'), ('valve', 'NN'), ('pulmonary', 'JJ'), ('intracardiac', 'NN'), ('lipomatous', 'JJ'), ('interatrial', 'JJ'), ('septum', 'NN')

An analysis is conducted on the POSTags column. A bar graph is created with the x-axis representing the various POS tags (Part-of-Speech tags), and the y-axis representing the frequency count of each POS tag.

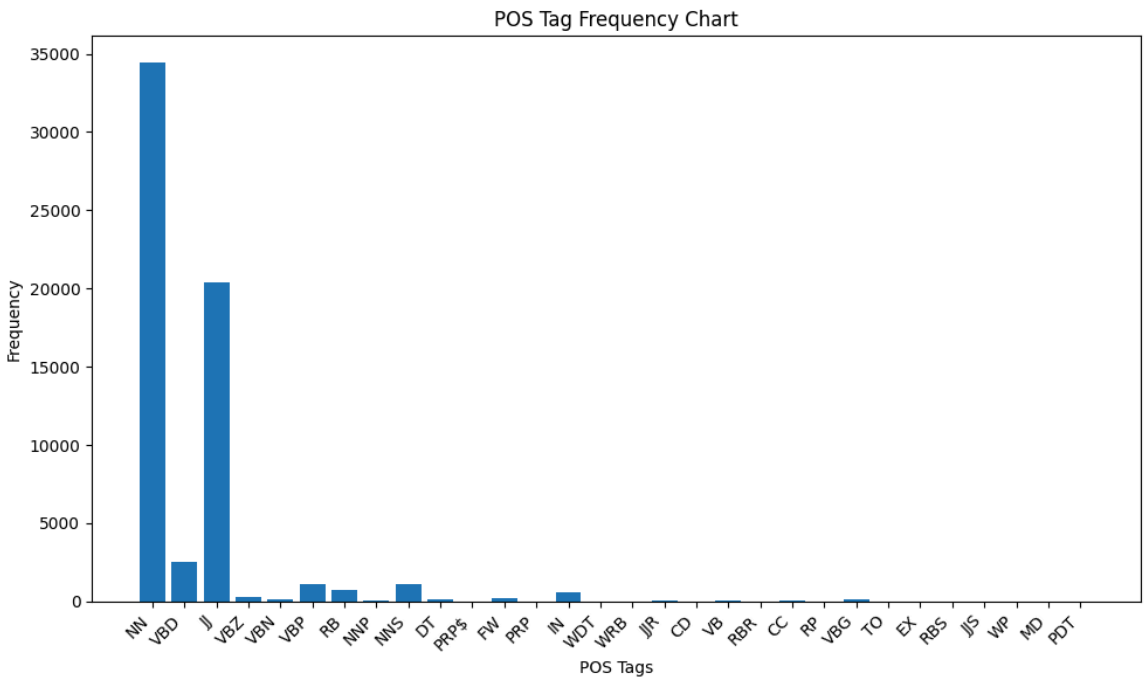

Figure 11: Bar chart showing the frequency of pos tags in the entire transcription

From the bar chart it is evident that NN is the most common pos tag because it shows the most frequency in the bar chart followed by JJ is which is just higher than the half of NN's frequency which makes it the second most common pos tag. It is evident from the bar chart that NN and JJ contribute a lot of words in the text.

The NN tag is used for nouns that can be singular or plural and may represent either a single object or a mass of something. whereas JJ is used to indicate adjectives, which are words that describe or modify nouns. Adjectives provide additional information about the attributes, qualities, or characteristics of nouns.

The most frequent pos tags other than NN and JJ are VBD, VBP and NNS all others

VBD is used for verbs that are in the past tense. In this context such words are considered not relevant so every VBP words are removed from the text.

But NNS is used for nouns that are in the plural form, indicating more than one instance of an object, which is more useful in this case,

suppose this is the text we have,

left ventricular wall wall left ventricular left ventricular left ventricular left atrial left atrium atrium ventricle aortic root aortic valve aortic valve aortic valve area mitral annular leaflet mitral valve mitral tricuspid valve tricuspid pulmonary artery pulmonary artery pulmonary valve pulmonary intracardiac lipomatous interatrial septum

Here,

NN are wall, wall, atrium, atrium, ventricle, root, valve, valve, valve, area, leaflet, valve, valve, artery, artery, valve, intracardiac, septum. VBD are left, left, left, left. JJ are ventricular, ventricular, ventricular, ventricular, atrial, left, aortic, aortic, aortic, aortic, mitral, annular,

mitral, mitral, tricuspid, tricuspid, pulmonary, pulmonary, pulmonary, pulmonary, lipomatous, interatrial and VBN left. There are no VBP and NNS in this text.

From this it is clear that VBD and VBZ are not relevant and also their frequency is very less too.

So, A new column called MedicalTerms is created with only the relevant postags which are NN, JJ and NNS.

The text in the 'MedicalTerms' column is stored as an array. This array is then joined to form a continuous text, which is subsequently moved to a new column called 'MedicalText.' From the resulting DataFrame, only the 'MedicalText' and 'medical_specialties' columns are retained, while all other columns are deleted to keep these two.

The resulting DataFrame consists of 1471 entries.

3.7. Feature engineering

Feature engineering involves manipulating a dataset to create informative variables, known as features, which are utilized to train a machine learning model for predictive tasks(Anon, n.d.). It refers to skillfully molding data to leverage the inherent tendencies of the machine learning method, ideally enhancing both accuracy and interpretability(Verdonck et al., 2021). The objective of feature engineering is to devise intelligent features through either modifying current ones using diverse conversions or generating fresh, relevant features from varied origins.

3.7.1. Vectorisation

Our feature engineering is initiated by importing vital libraries, notably including the TfidfVectorizer class from the renowned scikit-learn library. This class emerged as a pivotal tool in creating TF-IDF vector representations from our textual data. Notably, TF-IDF (Term Frequency-Inverse Document Frequency) is a powerful technique that enables the conversion of raw text into numerical forms suitable for advanced machine learning processes.

After Applying the TF-IDF vectorisation the shape becomes (1471, 1000).

The change in shape from (1471, 3) to (1471, 1000) after vectorisation indicates a transformation in the structure and content of your Data Frame due to the application of the TF-IDF vectorisation process.

The transformation from (1471, 3) to (1471, 1000) signifies that the textual data has been converted into numerical representations using TF-IDF vectorisation, resulting in a matrix where each entry's text has been transformed into a vector of 1000 numerical values. This numerical representation is more suitable for various machine learning tasks that enables to perform analyses, clustering, classification, and other tasks on the transformed data.

A t-SNE plot is plotted to visually represent the relationships between medical specialties in a lower-dimensional space (Figure 12).

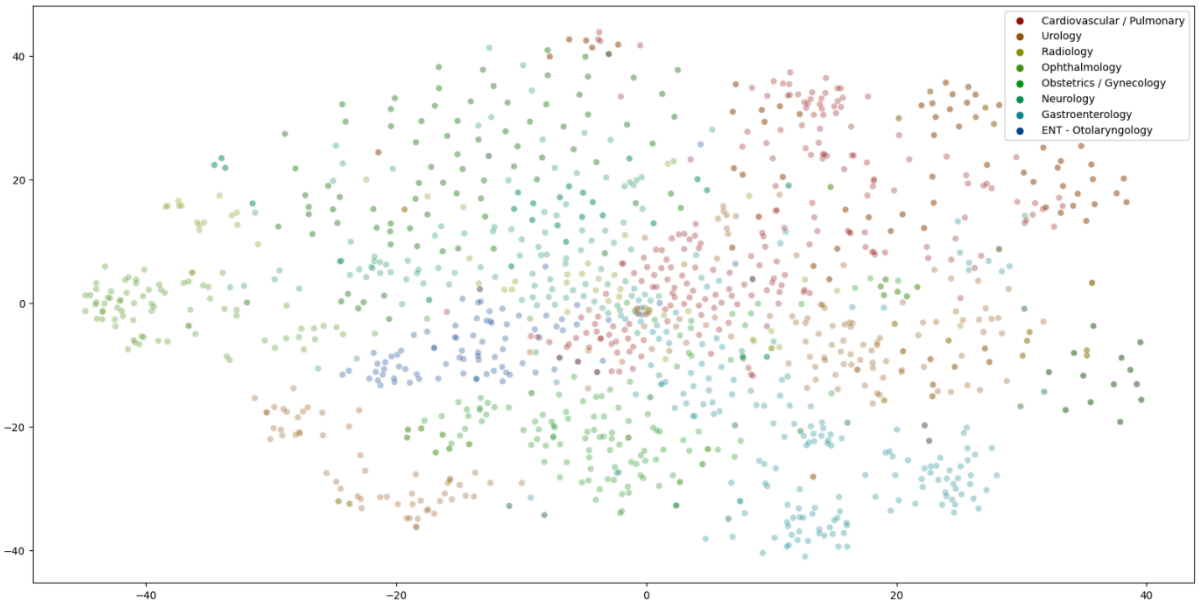

Figure 12: t-SNE plot of vectorised data

The plot shows some clusters but still some categories are overlapping each other.

3.7.2. Dimensionality reduction

Principal Component Analysis (PCA) is a dimensionality reduction technique commonly applied to vectorized data, aiming to transform it into a lower-dimensional space while retaining as much variance as possible.

This reduction in dimensions not only compresses the data and removes redundancies but also aids in noise reduction. The resulting lower-dimensional representation captures the essence of the original data, making it suitable for visualisation, analysis, and machine learning tasks.

To check the reduction in the dimensionality a t-SNE plot is plotted (Figure 14),

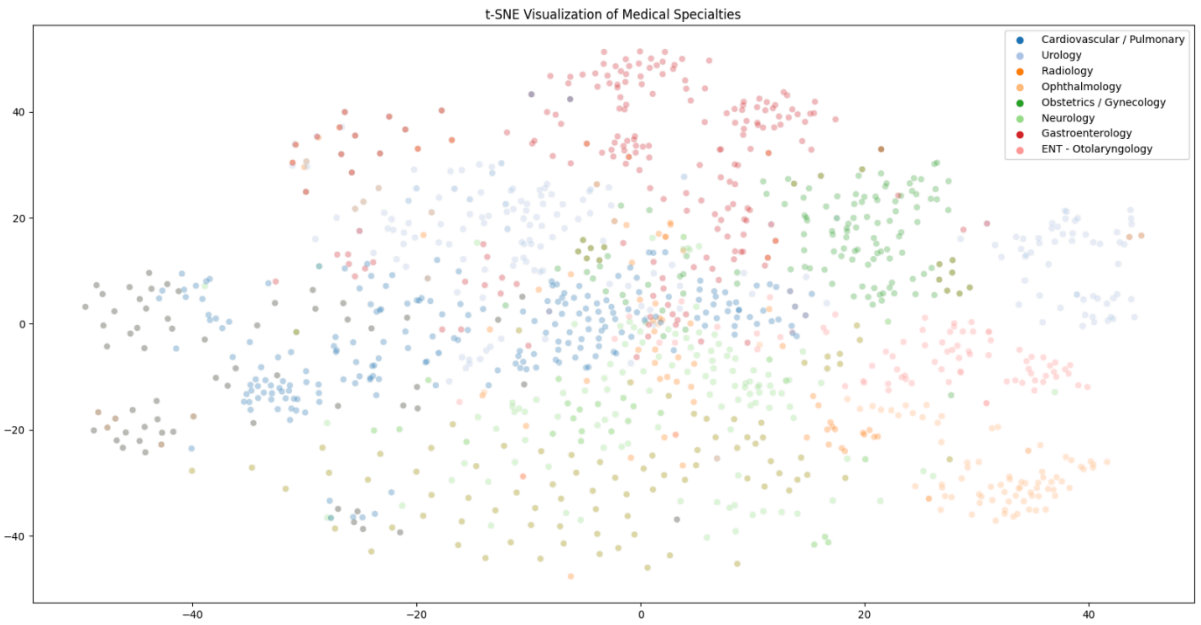

Figure 14: t-SNE plot after PCA

From this plot, it becomes evident that the clusters are more distinct and isolated compared to the previous plot obtained immediately after vectorization. The extent of overlap between clusters has significantly reduced, indicating that the dimensionality reduction techniques have effectively enhanced the separation of data clusters.

3.8. Class imbalance

The dataset is found to have class imbalance where some specialties are more in number and some are less in number.

In this next step, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to address class imbalance within the dataset. Class imbalance can occur when some classes have significantly fewer samples than others, potentially leading to biased model performance.

SMOTE technique is applied by generating synthetic samples for the minority class. The over-sampled data is then split into training and testing sets for subsequent machine learning tasks, ensuring that the class distribution is preserved in the splits. This approach enhances the model's ability to learn from underrepresented classes and improve overall performance.

A bar plot visualisation is created to understand the impact that SMOTE made on the data (Figure 15).

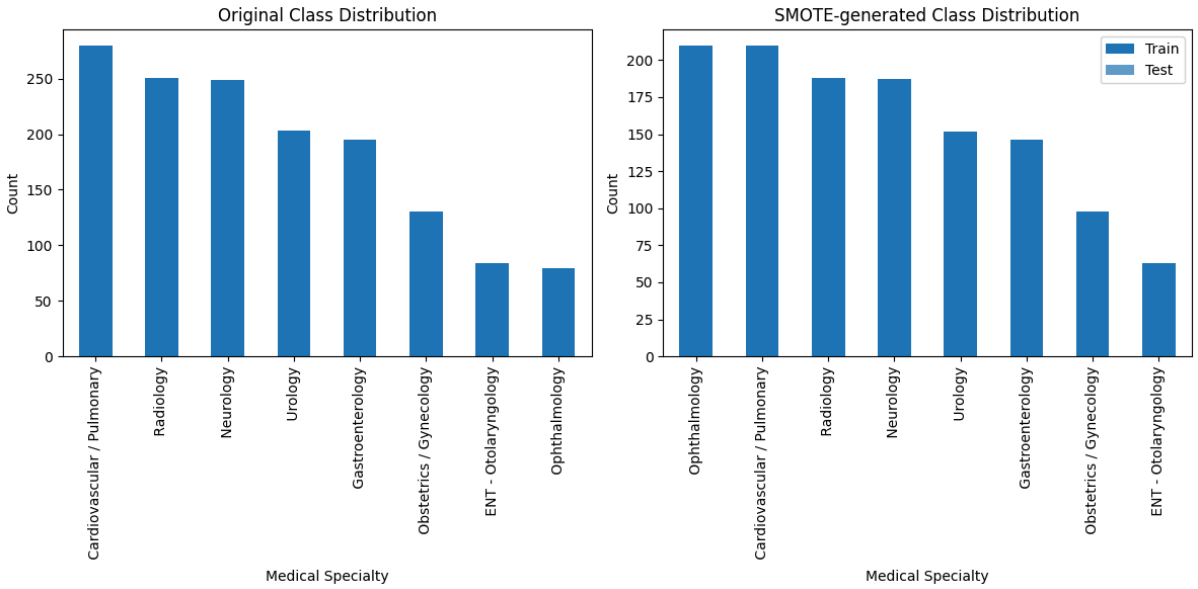

Figure 15:Count of specialties before and after applying SMOTE

The bar plot visualisation (Figure 15) and Figure 16 clearly illustrates the effective impact of SMOTE on the data. The original dataset had varying instance counts across medical specialties, with some like "Cardiovascular / Pulmonary," "Radiology," and "Neurology" having higher counts, and others like "Ophthalmology" and "ENT - Otolaryngology" having fewer. SMOTE was used to balance the training set, matching the highest-class count. This increased underrepresented classes' instances like "Ophthalmology" to be comparable. The balanced class distribution in the test set indicates consistent application of SMOTE for both training and testing.

```
Original class distribution:
 Cardiovascular / Pulmonary    280
 Radiology                     251
 Neurology                     249
 Urology                       203
 Gastroenterology              195
 Obstetrics / Gynecology       130
 ENT - Otolaryngology           84
 Ophthalmology                  79
```

```
SMOTE-generated class distribution in training set:
 Ophthalmology                 210
 Cardiovascular / Pulmonary    210
 Radiology                     188
 Neurology                     187
 Urology                       152
 Gastroenterology              146
 Obstetrics / Gynecology        98
 ENT - Otolaryngology           63
dtype: int64
SMOTE-generated class distribution in test set:
 Ophthalmology                 70
 Cardiovascular / Pulmonary    70
 Radiology                     63
 Neurology                     62
 Urology                       51
 Gastroenterology              49
 Obstetrics / Gynecology       32
 ENT - Otolaryngology          21
```

Figure 16: Original class distribution and smote generated class distribution

A X_train, X_test, y_train and y_test is created using train_test_split method from the sampled data.

3.9. Model building

Two machine learning algorithms are applied to the training data X_train and y_train to predict the medical specialties for the test data X_test. The predictions from these algorithms are then compared with the actual medical specialties in the y_test dataset. This comparison allows for the evaluation of the algorithms' performance in predicting the correct medical specialties.

Additionally, a user input interface is implemented, prompting the user to input text data, which represents a transcription. This input text is then processed through entity extraction followed by vectorisation and pca. Subsequently, the machine learning algorithm is employed to predict the medical specialty corresponding to this input. This entire process is designed to assess the algorithm's predictive accuracy and evaluate its performance in making medical specialty predictions based on user-provided text data.

3.9.1. Random Forest:

The Random Forest algorithm is imported from the sklearn.ensemble library which provides functionalities for ensemble learning methods. In this context, the Random Forest algorithm is applied to the dataset for classification tasks. After applying the algorithm, the accuracy metric is computed, revealing an accuracy value of 0.64 which indicates that the algorithm correctly predicted the medical specialties for approximately 64 percent of the test instances.

A classification report is generated to analyse the prediction accuracy and F1-score for each medical specialty. This report provides insights into how well the machine learning model performs for each specific specialty, helping to assess its overall predictive capabilities.

3.9.2. Logistic Regression:

The Logistic Regression model was imported from the sklearn.linear_model library, which provides implementations for various linear regression algorithms.

Logistic regression is a popular, versatile, easily understood algorithm that handles sparse data well, works for binary and multi-class classification, and serves as a strong baseline model for text classification(Otten, 2023).

Likewise, Logistic Regression is applied to the training data, resulting in an accuracy of 0.82.

The subsequent classification report shows the F1 scores for each speciality as well.

The flow chart below shows the whole steps involved in the process:

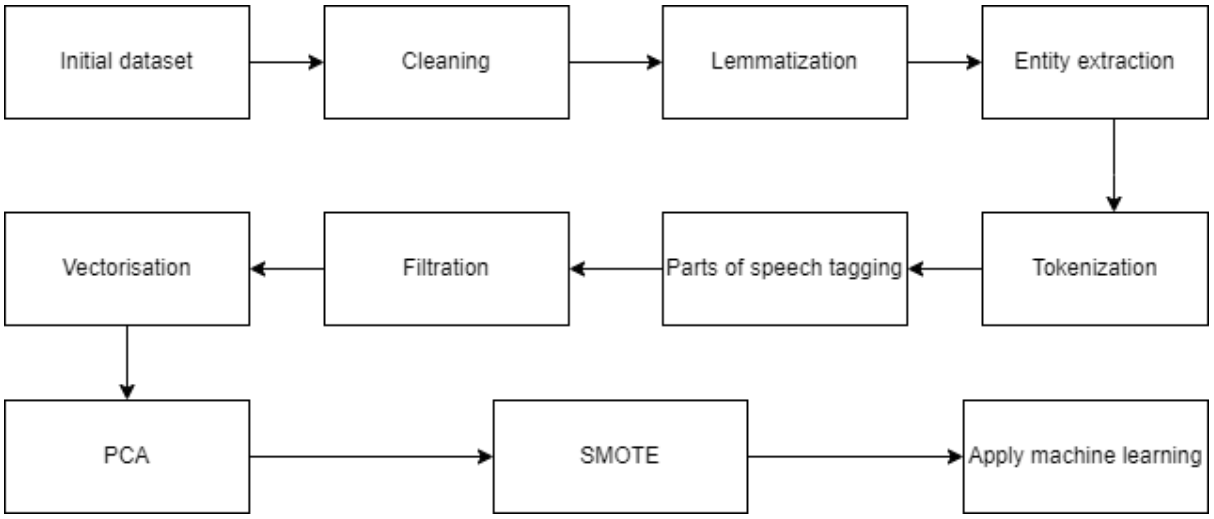

Figure 17:Flow chart of entire process involved in the process

4. RESULT AND FINDINGS

4.1. Model Comparison

Random Forest Model Evaluation:

Classification Report For Random Forest:

	precision	recall	f1-score	support
Cardiovascular / Pulmonary	0.57	0.74	0.64	70
ENT - Otolaryngology	0.88	0.71	0.79	21
Gastroenterology	0.82	0.65	0.73	49
Neurology	0.53	0.56	0.55	62
Obstetrics / Gynecology	0.92	0.72	0.81	32
Ophthalmology	1.00	0.96	0.98	70
Radiology	0.11	0.11	0.11	63
Urology	0.78	0.75	0.76	51
accuracy			0.64	418
macro avg	0.70	0.65	0.67	418
weighted avg	0.66	0.64	0.65	418

Figure 18: Classification report of logistic regression

The Random Forest model achieved an overall accuracy of 64 percentage. It performed well in predicting specialties such as Ophthalmology, Obstetrics / Gynaecology with high f1 score ranging from 0.81 to 0.98 whereas Urology, ENT – Otolaryngology, cardiovascular/pulmonary, gastroenterology shows medium performance. However, the model faced challenges in predicting specialties like "Radiology" and "Neurology" where the F1-scores were relatively lower (ranging from 0.11 to 0.55). The macro-average F1-score for this model is 0.67, indicating a moderate overall performance.

Logistic Regression Model Evaluation:

```
Classification Report for Logistic Regression:
                              precision    recall  f1-score   support

 Cardiovascular / Pulmonary       0.74      0.79      0.76        70
       ENT - Otolaryngology       1.00      0.81      0.89        21
           Gastroenterology       0.86      0.78      0.82        49
                  Neurology       0.56      0.61      0.58        62
    Obstetrics / Gynecology       0.93      0.78      0.85        32
               Ophthalmology       0.97      0.97      0.97        70
                  Radiology       0.22      0.22      0.22        63
                    Urology       0.78      0.82      0.80        51

                   accuracy                           0.71       418
                  macro avg       0.76      0.72      0.74       418
               weighted avg       0.72      0.71      0.71       418
```

Figure 19: Classification report of Random Forest

The Logistic Regression model achieved an overall accuracy of 71%. It exhibited strong performance in predicting specialties such as Ophthalmology, ENT – otolaryngology, Gastroenterology and Obstetrics / Gynaecology with high F1 scores ranging from 0.82 to 0.97. However, it displayed more moderate performance in classes like Cardiovascular/Pulmonary and Urology, with F1 scores ranging from 0.76 to 0.80. Challenges were observed in predicting classes such as Radiology and Neurology, where the F1-scores were relatively lower (ranging from 0.22 to 0.58). The macro-average F1-score for this model is 0.74, indicating a balanced overall performance across classes.

In comparison, the Logistic Regression model demonstrated enhanced predictive performance with a higher overall accuracy and improved F1-scores for most specialties. Both models show strengths in certain areas while encountering difficulties in others, and this evaluation provides insights into their respective capabilities and limitations.

4.2. Research questions and findings:

1. What is the comparative performance of different machine learning algorithms in classifying medical transcriptions into their respective specialties, and how does this performance vary based on the pre-processing steps applied?

The comparative performance of different machine learning algorithms in classifying medical transcriptions into their respective specialties varies based on the pre-processing steps applied. The Random Forest model achieved an overall accuracy of 64%, while the Logistic Regression model exhibited an improved accuracy of 71%.

Both models performed well in predicting certain specialties such as ENT - Otolaryngology, Gastroenterology, Obstetrics / Gynaecology, and Ophthalmology, where precision, recall, and F1-scores were relatively high. However, they faced challenges in predicting other specialties like Cardiovascular / Pulmonary, Neurology, and Radiology, where the F1-scores were relatively lower.

Comparing the macro-average F1-scores, the Logistic Regression model exhibited a higher score of 0.74, indicating a relatively balanced overall performance compared to the Random Forest model's macro-average F1-score of 0.67.

It's evident that the choice of machine learning algorithm, along with the pre-processing steps, plays a significant role in the classification performance. While both models demonstrated strengths and weaknesses in predicting various medical specialties, the Logistic Regression model showed an edge in accuracy and overall balance. This suggests that the Logistic Regression model, coupled with appropriate pre-processing, could be more suitable for classifying medical transcriptions into their respective specialties in this context.

2. How does text mining impact medical data analysis and contribute to simplifying the process?

Text mining significantly impacts medical data analysis by enhancing the efficiency and effectiveness of processing large volumes of textual medical information. The initial transcription was quite noisy, but the application of text mining reduced the noise and made it more amenable for the application of machine learning.

Lemmatization ensures that words are reduced to their base forms, reducing variations and improving the accuracy of subsequent analysis. Entity extraction, specifically with a bio NER model, enables the identification of medical terms, conditions, and entities, enriching the data with contextually relevant information.

Tokenization breaks down text into meaningful units, such as words or phrases, which facilitates a better understanding of the context. This is particularly crucial in medical data analysis where precise interpretation is paramount.

These methods help to filter out unwanted or noisy data, to eliminate irrelevant information that could introduce confusion or bias into the analysis. This ensures that the remaining data is more focused and meaningful.

automate processes that would be time-consuming and error-prone if done manually. This automation leads to quicker insights and enables more efficient utilization of resources.

Text mining simplifies the process by handling complex tasks like identifying entities and normalizing words, which would require significant cognitive effort if done manually. This allows analysts to focus on higher-level interpretation and decision-making.

The initial transcription was very noisy but application of text mining reduces the noise and made it easy for applying machine learning.
In the healthcare sector, precision and accuracy hold paramount importance due to the critical nature of decisions made based on classifications. While achieving a 71 percent accuracy through Logistic Regression is a positive step, the unique sensitivity of the health domain demands the need for even higher accuracy levels. The potential consequences of misclassification in medical contexts necessitate more advanced techniques to pre-process the data. Using the scispaCy model significantly reduced the noise in the data and could extract the medical entities However, it's essential to acknowledge that while text mining plays a vital role, accurate classification demands more than this singular approach. To enhance classification accuracy, a combination of feature engineering, domain-specific

knowledge integration, and potentially more advanced techniques remain crucial components in predicting accurate class assignments.

3. What are the methods to improve the accuracy of classification?

Collaborating with domain experts can provide insights into nuances and patterns that algorithms might miss. Their expertise can guide feature selection, model design, and interpretation of results.

Exploring more advanced AI models, such as GPT-4, could enhance accuracy. GPT-4's contextual understanding can assist in fine-tuning classification based on complex language and context(Liu et al., 2023).

Utilize ensemble techniques like stacking or bagging to combine multiple models' predictions, benefiting from their diverse strengths and mitigating individual model weaknesses.

Establishing a more structured and standardized process for collecting medical transcriptions can help reduce noise. Clear guidelines for transcription methods and terminology can lead to more consistent and reliable data. Collaborate with medical institutions to encourage proper and comprehensive recording of medical transcriptions. Sharing the potential benefits of accurate data collection can motivate institutions to improve their practices.

Challenges:
Some of the challenges faced by researchers in this area include data heterogeneity, the presence of domain-specific language, large label space, and the need for effective validation and evaluation methods(Gonzalez et al., 2012).

5. CONCLUSION

This project explored the use of text mining and machine learning techniques to classify text. Text mining techniques were used to extract features. These features were then used to train machine learning models, such as logistic regression, to classify text.

The results of the project showed that text mining and machine learning can be used to accurately classify text. However, there are still challenges to overcome, such as the evolving nature of language and the need to understand domain-specific terminology.

Despite these challenges, the project showed the potential of text mining and machine learning to interpret and leverage the latent insights within textual data. These methodologies can be used for knowledge discovery, informed decision-making, and the realization of data's transformative capability.

BIBLIOGRAPHY:

Abdel-Fattah, M.K. et al. (2021) Quantitative Evaluation of Soil Quality Using Principal Component Analysis: The Case Study of El-Fayoum Depression Egypt. *Sustainability*, 13(4), p. 1824.

Averaged Perceptron Tagger. Available from : https://www.kaggle.com/datasets/nltkdata/averaged-perceptron-tagger [Accessed 30/08/23a].

Breiman, L. (2001) Random Forests. *Machine Learning*, 45(1), pp. 5–32.

Chandrayan, B. (2021) A brief survey of Text Mining and its applications. *International Journal of Emerging Trends in Engineering Research*, 9(8), pp. 1190–1195.

Chen, S. (2020) *Getting Started with Text Vectorization*. [Online] Medium. Available from : https://towardsdatascience.com/getting-started-with-text-vectorization-2f2efbec6685 [Accessed 30/08/23].

David, G.C. et al. (2009) Listening to what is said - transcribing what is heard: the impact of speech recognition technology (SRT) on the practice of medical transcription (MT). *Sociology of Health & Illness*, 31(6), pp. 924–938.

Denisko, D. and Hoffman, M.M. (2018) Classification and interaction in random forests. *Proceedings of the National Academy of Sciences of the United States of America*, 115(8), pp. 1690–1692.

Divakaran, M. (2019) *scispaCy for Bio-medical Named Entity Recognition(NER)*. [Online] Medium. Available from : https://maheshdivakaran.medium.com/scispacy-for-bio-medical-named-entity-recognition-ner-63ed548f1df0 [Accessed 30/08/23].

Gong, D. (2022) *Top 6 Machine Learning Algorithms for Classification*. [Online] Medium. Available from : https://towardsdatascience.com/top-machine-learning-algorithms-for-classification-2197870ff501 [Accessed 30/08/23].

Gonzalez, G. et al. (2012) TEXT AND DATA MINING FOR BIOMEDICAL DISCOVERY. In: *Biocomputing 2013*. Proceedings of the Pacific Symposium. Kohala Coast, Hawaii, USA: WORLD SCIENTIFIC, pp. 368–372.

Green, 1400 Crescent, Cary, S. 210 and Nc 27518 919.459.4600 *Improved Patient Care with Text Analytics & NLP*. Available from : https://www.zencos.com/blog/text-mining-nlp-healthcare-insights/ [Accessed 30/08/23].

Keerthi Vasan, K. and Surendiran, B. (2016) Dimensionality reduction using Principal Component Analysis for network intrusion detection. *Perspectives in Science*, 8, pp. 510–512.

Khyani, D. and B S, S. (2021) An Interpretation of Lemmatization and Stemming in Natural Language Processing. *Shanghai Ligong Daxue Xuebao/Journal of University of Shanghai for Science and Technology*, 22, pp. 350–357.

Kumar, L. (2013) TEXT MINING: CONCEPTS, PROCESS AND APPLICATIONS. *Journal of Global Research in Computer Science*, Vol. 4, pp. 36–39.

LessenText, A.P. from (2023) *Text Summarization using TF-IDF*. [Online] Medium. Available from : https://towardsdatascience.com/text-summarization-using-tf-idf-e64a0644ace3 [Accessed 30/08/23].

Liu, Y. et al. (2023) Summary of ChatGPT-Related Research and Perspective Towards the Future of Large Language Models. *Meta-Radiology*, 1(2), p. 100017.

Mansouri, A., Affendey, L.S. and Mamat, A. (2008) Named Entity Recognition Approaches.

Mathuna, K.T. and Shanthi, I.E. (2021) Text Mining in Health Care Sector Techniques, Applications and Challenges. In: *2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS)*. 2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS). pp. 1539–1545.

Mckinney, W. (2011) pandas: a Foundational Python Library for Data Analysis and Statistics. *Python High Performance Science Computer*.

Medical Transcription: Process, Guidelines, Notes | *OneDatai*. (2022) Available from : https://onedatai.com/medical-transcription-process-guidelines-notes/ [Accessed 30/08/23].

Mehta, J. et al. (2018) Use of Medical Transcription Data for Real World Clinicogenomic Evidence Generation for Mental Health Disorders In US. *Value in Health*, 21, p. S192.

Mohan, V. (2016) Text Mining: Open Source Tokenization Tools: An Analysis. 3, pp. 37–47.

Morgenthaler, S. (2009) Exploratory data analysis. *WIREs Computational Statistics*, 1(1), pp. 33–44.

Narayanan, S. (2013) Feasibility of Use of Medical Transcription Data for Real World Evidence Generation in the United States: A Pilot Study of Patients with Systemic Lupus Erythematosus Initiated on Belimumab. *Value in Health*, 16(7), p. A584.

National University of Computer and Emerging Sciences, Faisalabad, 38000, Pakistan et al. (2017) Data Cleaning In Data Warehouse: A Survey of Data Pre-processing Techniques and Tools. *International Journal of Information Technology and Computer Science*, 9(3), pp. 50–61.

Otten, N.V. (2023) *How To Implement Logistic Regression Text Classification [2 Ways]*. [Online] Spot Intelligence. Available from : https://spotintelligence.com/2023/02/22/logistic-regression-text-classification-python/ [Accessed 31/08/23].

Pagad, N.S. et al. (2022) Clinical Text Data Categorization and Feature Extraction Using Medical-Fissure Algorithm and Neg-Seq Algorithm. *Computational Intelligence and Neuroscience*, 2022, p. e5759521.

Pai, A. (2020) *What is Tokenization in NLP? Here's All You Need To Know*. [Online] Analytics Vidhya. Available from : https://www.analyticsvidhya.com/blog/2020/05/what-is-tokenization-nlp/ [Accessed 30/08/23].

Papers with Code - Feature Engineering. Available from : https://paperswithcode.com/task/feature-engineering [Accessed 30/08/23b].

Perera, N., Dehmer, M. and Emmert-Streib, F. (2020) Named Entity Recognition and Relation Detection for Biomedical Information Extraction. *Frontiers in Cell and Developmental Biology*, 8, [Online] Available from: https://www.frontiersin.org/articles/10.3389/fcell.2020.00673 [Accessed 30/08/2023].

Plisson, J., Lavrac, N. and Mladenic, D. A Rule based Approach to Word Lemmatization.

Python | Lemmatization with NLTK. (2018) [Online] GeeksforGeeks. Available from : https://www.geeksforgeeks.org/python-lemmatization-with-nltk/ [Accessed 30/08/23].

Rahman, M.M. and Davis, D.N. (2013) Addressing the Class Imbalance Problem in Medical Datasets. *International Journal of Machine Learning and Computing*, pp. 224–228.

Ramachandran, R. and Arutchelvan, K. (2021) Named entity recognition on bio-medical literature documents using hybrid based approach. *Journal of Ambient Intelligence and Humanized Computing*, [Online] Available from: doi.org/10.1007/s12652-021-03078-z [Accessed 30/08/2023].

Riddick, G. et al. (2011) Predicting in vitro drug sensitivity using Random Forests. *Bioinformatics*, 27(2), pp. 220–224.

Rosário-Ferreira, N. et al. (2021) The Treasury Chest of Text Mining: Piling Available Resources for Powerful Biomedical Text Mining. *BioChem*, 1(2), pp. 60–80.

Shlens, J. (2014).

Spoustová, D. "johanka" et al. (2009) Semi-Supervised Training for the Averaged Perceptron POS Tagger. In: *Proceedings of the 12th Conference of the European Chapter of the ACL (EACL 2009)*. EACL 2009. Athens, Greece: Association for Computational Linguistics, pp. 763–771.

Talib, R. et al. (2016) Text Mining: Techniques, Applications and Issues. *International Journal of Advanced Computer Science and Applications*, 7(11), [Online] Available from: doi.org/10.14569/IJACSA.2016.071153 [Accessed 30/08/2023].

Transcribed Medical Transcription Sample Reports and Examples - MTSamples. Available from : https://mtsamples.com/?utm_content=cmp-true [Accessed 30/08/23c].

Verdonck, T. et al. (2021) Special issue on feature engineering editorial. *Machine Learning*, [Online] Available from: doi.org/10.1007/s10994-021-06042-2 [Accessed 30/08/2023].

What is Logistic regression? | IBM. Available from : https://www.ibm.com/topics/logistic-regression [Accessed 30/08/23d].

APPENDIX

DE MONTFORT UNIVERSITY LEICESTER

Research Ethics Screening Questions

This form will help you establish whether an application for research ethics approval is required. If you are unsure, please discuss with your research supervisor or Faculty Research Ethics Committee.

Name: Sidharth Vengathattil
Project Title: Classification of Medical Transcription using Text mining and Machine learning

Screening

PART 1: Screening Question	Further Guidance	Yes	No
Does your research involve accessing, storing or disseminating security sensitive information such as that relating to extremism or terrorism?	For further details, see DMU's policy on security sensitive research.	☐	☒

If' yes', please contact your Faculty Research Ethics Committee (see below)
If 'no', move onto Part 2.

PART 2: Screening Question	Further Guidance	Yes	No
Does your research involve only secondary data that are: a. Curated for public access, or b. Data for which consent was provided for reuse in research beyond that for which it was originally collected, and those data are anonymized?	Examples of data already in the public domain include those curated for public access and display or for re-use by researchers, material held in archives or private collections. Refer to the Research Ethics Code of Practice for further details regarding secondary data.	☒	☐
Is your project exclusively based on published literature or library/archival materials which have been specifically curated for general public access or display?	Examples include literature reviews, meta-analysis or re-use of published data.	☒	☐

If 'yes' to any of the questions in Part 2, an ethics application IS NOT required. See actions below:

- **Staff:** Please retain a copy of this form in your project file.
- **Post-graduate research students:** Please confirm the declaration at the end of this form. Your supervisor will then need to forward the form to the Faculty Research Ethics Committee and Doctoral College.
- **Undergraduate/Taught Postgraduate students:** Please confirm the declaration at the end of this form. Your programme team will inform you what to do next.

If 'no' to both of the above questions, please proceed to Part 3 (next page).

DE MONTFORT UNIVERSITY LEICESTER

Research Ethics Screening Questions

PART 3: Screening Question	Further Guidance	Yes	No
Does your research involve secondary data not covered by Part 1?	This may include identifiable data that are not in the public domain, or where consent was not provided for resuse of data beyond the purposed for which it was originally collected.	☐	☒
Will your research involve human participants?	This includes any procedures or activities (including questionnaires, surveys, interviews and focus groups) involving living participants and the use of data or tissue obtained from living individuals, whether directly or indirectly. This includes data about people obtained from databases, public websites and social media, unless held in publicly available collections or from archaeological sites and curated explicitly for the purposes of research. This may also include the use of data and tissue obtained from deceased individuals. Tissue (or material derived from human tissue e.g. cell lines) which has been obtained from commercial sources will not normally require ethical review. Please seek advice from ethics@dmu.ac.uk.	☐	☒
Will your research involve the use of animals?	This includes any research involving live animals (e.g. intervention, manipulation of conditions or observation) and animal tissue. Tissue (or material derived from animal tissue e.g. cell lines) which has been obtained from commercial sources will not normally require ethical review.	☐	☒
Does this project involve the use of sensitive or restricted materials?	Please refer to the Sensitive Research Policy. Research covered by this policy includes: • Research into illegal activities, including the collection of source data, e.g. crime statistics; • Research which requires access to web sites normally prohibited on university servers; including, but not limited to; pornography, or the sites of any of the organisations proscribed by the UK Government. • Research into extremism and radicalisation.	☐	☒
Does the project have the potential to cause environmental damage or harm?	This includes the use of equipment, materials, techniques or research results that could have an impact on the environment, ecology, hydrology or on local communities during or after the research project.	☐	☒
Does your project require external ethical review?	External review will be required for research involving NHS patients, staff, data or facilities (includes contractors providing services under contract with care services or commissioners). Further examples include: • UK Social Care organisations or service users (includes contractors providing services under contract with care services or commissioners). • Clinical Trials of Investigational Medicinal Products, Medical Devices or Gene Therapy Medicinal Products. • The prison service, offenders or participants on probation. Any member of the research team or participants who are members of the Armed Forces or their entitled dependents. • The administration of ionising radiation. • Participants aged 16 or over who may lack capacity to give fully informed consent or who may lose capacity to give	☐	☒

	fully informed consent during the study. • Research regulated under the Animals (Scientific Procedures) Act 1986 (ASPA). • Research that may be regulated under the Human Fertility & Embryology Authority Please seek advice from ethics@dmu.ac.uk before proceeding any further if your research involves any of the above.		

If 'yes' to any of the Part 2 questions, an ethics application IS required.
Please submit your application using the WorkTribe ethics module.

Declaration (for students use, where required by their programme)

I take full responsibility for the information provided in this application, and: a. An ethics application IS required. I confirm I will not commence research before receiving a favorable ethics opinion, OR; b. I am satisfied that an ethics application is NOT required. I have sought the advice as necessary, and will monitor my research as it progresses and reassess the need for an ethics application periodically. If you cannot confirm the above, please contact your Faculty Research Ethics Committee.	☒
As the student's supervisor, I am satisfied with the outcome of this assessment.	☒

Faculty Research Ethics Committees:

Faculty of Business and Law (BAL)	BALResearchEthics@dmu.ac.uk
Faculty of Arts, Design and Humanities (ADH)	adhethics@dmu.ac.uk
Faculty of Computing, Engineering and Media (CEM)	ethics.CEM@dmu.ac.uk
Faculty of Health and Life Sciences (HLS)	hlsfro@dmu.ac.uk